

FINAL SUMMARY (WITH YOUR REAL VALUES)

Customer Churn Prediction & Risk Segmentation

Objective:

The objective of this project is to predict customer churn and segment customers into risk categories to enable proactive and data-driven retention strategies.

Approach:

- Performed data cleaning and preprocessing, including handling missing values and encoding categorical variables
 - Engineered meaningful features such as **ChargesPerMonth** and customer risk indicators
 - Trained and compared three models: Logistic Regression, Random Forest, and Gradient Boosting
 - Evaluated models using Accuracy, Precision, Recall, F1-score, and ROC-AUC
 - Implemented probability-based **customer risk segmentation** into High, Medium, and Low risk groups
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Key Insights:

- **Contract type is the strongest driver of churn:**
 - Month-to-month customers have a churn rate of **42.71%**
 - One-year contracts: **11.27%**
 - Two-year contracts: only **2.83%**
 - Customers who churn have significantly **lower average tenure (17.98 months)** compared to retained customers (**37.57 months**)
 - Customers who churn pay higher average monthly charges (**₹74.44**) compared to non-churned customers (**₹61.27**)
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Risk Segmentation:

- **Low Risk** 🟢: 71.61% of customers — stable and loyal
 - **Medium Risk** 🟡: 20.58% — potential churners requiring monitoring
 - **High Risk** 🔴: 7.81% — high likelihood of churn, immediate action required
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Business Recommendations:

1. Offer **discounted long-term contracts** to reduce churn among month-to-month customers, as they show the highest churn rate (42.71%)
 2. Implement **early-stage retention strategies** targeting customers within the first 6–12 months, as churn is higher among low-tenure users
 3. Provide **personalized pricing or loyalty benefits** for high-paying customers to reduce churn risk
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Limitations & Future Improvements:

- The dataset lacks behavioral and usage-based features such as customer complaints, service quality, and engagement metrics
- Model performance can be improved with advanced techniques such as **hyperparameter tuning, XGBoost, or ensemble models**
- Future work can include **real-time churn prediction systems** and integration with business dashboards